

ActInf Livestream #037.2 ~ "Free Energy: A User's Guide"

Livestream | 1:53:28

All right. Hello everyone. Welcome to Actimflab livestream number 37.2. It's February 9th, 2022. Welcome to the Active Inference Lab. We are a participatory online lab that is communicating, learning, and practicing applied active inference. You can find us at the links here on this slide. This is a recorded and an archived livestream, so please provide us with feedback so that we can improve our work. All backgrounds and perspectives are welcome here and we'll be following good video etiquette for livestreams. Feel free to visually raise your physical hand if you'd like, or use any of our emoji affordances. Check out activeinference.org to get updates on what the lab is up to and also get more information on how to participate, or check out this CODA link if you'd like to see our past and upcoming streams. We're here today in stream number 37.2, It's our third stream after the .0 background video and the .1 group discussion we had on the paper, Free Energy, A User's Guide by Mann, Payne, and Kirchhoff from 2021. And today we have a few things written. There's a couple sections of the paper that we didn't get to last time, and I'm sure everyone has brought some interesting thoughts. And of course, if you're watching live, please feel free to add any questions or comments into the live chat because we'll definitely have time to address anything there. We'll start with an introduction and warmup. So we'll say hi, mention anything that we're excited to talk about today, something that we liked or remembered or reviewed about the paper this past week, anything that we're wondering about. And as we just mentioned before we began, like, what does a .2 for a guide look like? So I'm Daniel, I'm a researcher in California, and I'm going to think while others are speaking and then return to what the .2 can be. And I'll pass it to Stephen.

Hello, I'm Stephen. I'm in Toronto. Stephen Sillit. I'm doing a practice-based PhD around social topographies and how they can be used in community development and theatre for development. And yeah, I'm looking forward

to looking at some of the implications of how this might be rolled out in different types of contexts.

I'll pass this over to Stephen. Stephen Sillit. I'm Dean. I'm in British Columbia at the moment. And as with all things guide related, I'm always interested in how we translate from the word to the deed, getting from a place of information that's curated to actions that are also maybe curated based on that bit of information. And I'll pass it to Blue.

Sorry, I lost my mute button. I'm Blue. I'm a research assistant or research consultant in New Mexico, transported back to my grad school days. I had like a mute button loss, a time warp all at the same time. I'm excited to get into the sections of the paper that we didn't really discuss, like in particular, like using mathematics to validate empirical claims or justify empirical claims. And also kind of zooming out and evaluating this work from the perspective of a guide. And like, I think about the Hitchhiker's guide and like, don't panic. And like, what are the things that we really like need? What are the essential components of the FEP and active inference that we need to communicate to others? And I'll pass it back to Daniel, who's going to tell us what a .2 really is.

Well, what it really has been is the

the far side of the bathtub. The .0, we're kind of dropping into the skating rink. And then in the .1, there's an opening and we explore. And last week was an awesome discussion. We got to speak with the first author. And now we continue, but also recollect some of the threads that we opened and build inertia towards where we take it next. And I think there's almost no genre more appropriate for a .2 than the guidebook. Like you put down in Dean's car manual metaphor, like you don't hold that up when you're driving. And so papers, similarly, they engage our regime of attention transiently. You're not supposed to only be looking at the paper. So we're going to put this paper down. We're going to end this video eventually. And then what will be left with us, how will our generative model have been updated so that our policy selections are different, whether they're mental action policies related to attention or metacognition or whether they're physical action policies. So I guess it relates to that quintessential .2 question of moving from the word of the paper that we're engaging in in our discussions to the deeds. And what are those deeds? And then it's almost like this is an action-oriented framing of a .2 or of just what we're up to today. And then there's this asynchronous or not temporally binding elements of, which is raised by the paper though, of what are the essential components of FEP and ACT-IMP, and also the differentiation perhaps between the conceptual and the pedagogical. Like are the pieces that are the minimal kernel of FEP and ACT-IMP, is that where we, is that page one on the guide? How does education, which is going to be the trajectory of a learner, relate to the theoretical apparatus, which is kind of like a building or some sort of structure that doesn't necessarily have a trajectory like format? So maybe just to return to this question of the word and the deed.

In the .1, we heard about what the target audience was from the author's perspective of the author's perspective for this paper, which was, it's a special issue. It's a topical collection called the Free Energy Principle from Biology to Cognition. And we heard a little bit about how the editor made a request of the authors to write this special piece. And they also summarized it in their last aim. We aim to increase philosophical understanding of active inference so that it may be more readily evaluated. So within the philosophical domain so that it can be evaluated. But what other domains or ideas are important here? And then what are these actions? So philosophers might be interested in reading this so that they can better philosophically evaluate. But what are the idea to deed pipelines that we're interested in or opening up here? Stephen, and then anyone else?

One of the aspects that this paper talks about is how much things can be screened off, how much it gives us latitude to separate out realistic empirical interpretations from the modeling of.

And I think that's the feature. They also talk about, and I'll be interested in what others think about this, but they talk about the Markov blanket as given this opportunity to screen or screen out what's inside and outside. And I wonder how this idea of screening, because screens are not the same as a wall. It's not, it's not walling off. But as come up on the Twitter conversation with Casper Hesp this week, this idea of leakage, like if I have a screen, I've got a blind here, as I described earlier, and there's some leakage through the blind, right? But is even that idea of screening off a little bit passive? I'm curious about that. A little bit dead. Anyway.

Yeah, if there weren't statistical dependencies between internal and external states, like if there were two states that

were just not even, there was no path from here to there, there wouldn't even be able to be blanket states partitioned out.

So internal and external states are conditionally independent upon the blanket states using the narrow pearl 1988 sense of Markov blanket. But there is a channel, there is a path from the internal to the external states.

Again, otherwise, they wouldn't even be part of the same click in that part of the Bayesian graph.

And then also, yes, Stephen?

Yeah, so that this, this, this, this click, this question of how much we need to find purchase through that blanket, and the screening off and how much we can get away with more broadly thinking about the free energy principle and inference and how that can be stepped up and down,

depending on where we are in terms of empiricism and trying to make claims.

Themes to feature a lot in this paper. And I'm, I'm, I'm, I'm sensing that because they're, they're sort of arguing in a way that as long as you can get functional, functional correlations, then how you get at them, be that approximate, isn't necessarily about it being the real world that gives you the, maybe different ways to arrive at something that's approximately

useful. And I think that speaks to dynamical systems over traditional systems in the sense that

I can have a system, which isn't the real system, but if my system flips or the regime shifts in similar ways to the target, as opposed to it actually being in and, and, and, and having the right types of values in it, in itself, it's more about what's the, what's the way that I start to see my uncertainty flip from one belief to another. If the, if the way the belief updating happens is still approximately useful enough for something to happen that's viable, then that is still okay.

And I'm wondering if that isn't possible in normal systems work, which isn't dynamical systems work.

Let's look at the order and kind of trace the trails of the authors. So in the early example, we had just the case of perceptual inference, which was first introduced in the exact Bayes context, and then we moved into the variational Bayes, catference, and we had W , world states, unobserved.

That's the actual temperature in the room. X , observed data, that's the reading on the thermometer.

So this is the simplest Bayesian latent or hidden state model. So is there a Markov blanket here?

Asking for a friend.

No, because there's only two nodes. So there cannot be a partitioning between internal and external states and the other states, conditioned on a third node. There's sort of a, it's kind of like the minimum two, but now we're taking to like minimum three. You can't have a blanket, an intervening or an intermediating or a state, which makes two others conditionally independent, unless you have three.

Three's a crowd, three's a party. Now in figure three, there's the basic model of action. So we still have the same definitions and meanings of W , unobserved world states, and X , observations.

Those observations are now being passed to the agent. And indeed they were implicitly being passed to the

agent above. If there would have been a third node agent there, it could have been said that the X was a Markov blanket, but as it's exactly written, there isn't. Now, here, what are the blanketing states?

X and Z. If you are conditioning upon X and Z, it makes W and the agents conditionally independent. And this is sort of presaging the Friston partition on the Markov blanket, which is to split up the notion of the total set of intermediating or conditionally independendizing states.

You know, Markov to Perl epoch and showing them in a way that's partitioned into incoming from the agent's perspective, statistical dependencies, which can be interpreted as sensory states or the observation of model data, and then outgoing statistical dependencies, Z, Z action you select. And then those can have some influence on world states.

Yes, Steven.

Steven R. You know, you say outgoing statistical dependencies, and that's also interesting because it's at some point the outgoing.

Cannot be statistical alone because you don't have the same kind of information availability.

So it tends to be more using energy and sort of least action type approaches.

So that would be, I'm sort of preempting that, but that's where it starts to split out later.

But they don't get bogged down in that at this stage.

They keep it all in the sort of Bayesian environment.

Not exactly sure what you mean, since these arrows represent not mechanistic or causal influence, but just statistical dependencies in the graph.

Steven R. Yeah, I was thinking more when that becomes action, when it's then translated to be action.

So then rather than sensory, which has much more sort of informational content, action could be seen as being more about what is more energy efficient, action or starts to look at the shape of the way that action happens.

So, but I'm a pretty interesting bit, but I'm wondering how, whether that is needed to be concerned about.

Steven R. If it were a software agent, then these are both informational, really.

And no matter what type of agent we're modeling, it still is just a statistical dependence.

Okay. Then you brought up questions about energy expenditure during action.

And that's one of the pieces that is sometimes challenging to parse out, which is like I'm first looking at variational free energy minimization.

So this is on the perceptual side. We're looking at the instantaneous fitting of a variational model of ongoing sensory information given the priors.

So that doesn't mean that for, for example, an organism, it's the lowest energy possible interpretation of the sensory data. And then when we bring action into the picture and look at expected free energy, neither does this mean that the Z that is selected is the laziest.

It just means that this G is being minimized. And so it's not the Gibbs free energy,

or the calories that are being minimized. It's something informational that has to do with model fitting. Yes, go for it.

Steven R. Yeah, no, that, that, that definitely makes sense.

And one thing Carl Friston mentioned in the symposium we did is this, this kind of two boundaries, there is kind of two boundaries happening in terms of Markovian boundaries. There's one coming in and there's one going out and the blankets kind of got both of those boundaries happening. So like you said, yeah, it's, it's the way that it then in the inference happens to know what actions to take is different, but the way that information is established is for sure different going through each of those. I mean, as it is with every mode of sensory data that comes in, because it could be vision, it could be hearing, it could be kinesthetic. So you've, you've got different types of input and you've got different types of output, but there, and this diagram at the moment is keeping both of those. There's no idea of leakage or, or those questions about how things leak across between, it's like, there's, there's two distinct flows going on, but that, and it simplifies it in that sense.

Yes. We've seen other topologies of perception, cognition, and action, like an edge connecting sense and action states or an arrow that's having a statistical dependency the other way. Okay. But this is the question for everybody here. So Steven, the first author talked about the fact that sort of the backstory of this whole guide thing was to, and I'm putting words into it, but I'm going to keep it reasonably short. I talked too much last week. He basically wanted to be able to remove some of the impediments and he pointed directly at some of the math that acted as a, maybe a kind of a, a bridge too far. And so he wanted to make that a simpler, um, maybe something that's more user friendly and therefore something that people will take up or adopt. And when you're putting on this, um, when you're putting up this slide here now, Daniel, I don't know whether, I know it eventually will be translated into the math formalisms. You just pointed, you had them up a minute ago, but I'm not really sure whether the evaluation aspect that, that philosophers are going to then decide whether something is worthy of their, their continued attention or not. I'm not sure if that, I don't know, how does this critical path, critical and evaluation being related in this context, how does this minimize or reduce the impediment nature of what the math is? Like if we're gonna, we're gonna guide somebody through this and their, and their, one of their hopes is to be able to say, okay, well, I'm not just making this stuff up. There's actually some, some way of being able to formalize this. How does that, through this kind of first step become, become easier, I guess, in terms of what the actual formula says? Because I know that when I first actually touched on the formulas myself, it didn't go back to a critical path.

Like there was no real physical example I could find in the world. And I think maybe that's what helped me understand that maybe this, this thinking or this ability to think statistically, sorry, to think statistically, maybe that requires, as Carl also said, our ability to do the math.

We have to be able to do the math and the math doesn't get simple. So I'm wondering what people think about that. Okay, Stephen, if you have a direct response. Yeah, yeah, well, I think that in some ways, this goes back to a little bit before where the math was, it helps us see why the math was put in at certain points by going to even a stage earlier. So I think there is some use in that, because I think,

like you say, there is a slight inflation in the way that the process theory then gets interpreted. And if it's already locked in too much in what's given to us with all the math, it's like, well, where did that math come from? And then that starts to get us involved in some of these earlier questions, because what's interesting here is the blanket, I think of the blanket often as a method for transfer.

It's a mode for transfer, just a bit like lithographic blankets transfer from an analog roller to a blanket roller, to a printed image and to the paper, it's transferring. Now, it also does mean that there's no direct contact between paper and paper and printing plate, right? So it's screened off as well. But it depends which way it's gone at, you know. And in this case here with these red circles, it's useful to think about, okay, well, there's something that's been inferred from the world. But that inference isn't directly accessible. It's not, that observed data isn't coming in as data, it's coming in as what's inferred to be about what's out there, which isn't the same as a signal. So it's like, how much is the agent able to have the environment transfer something onto them and transfer something onto the environment? And how much are they screened off from the environment? How much is the environment screened off from them? I suppose that's, that's something that relates to that.

Okay, thanks. So, Dean, you asked, how does the, yes, do you want to go for it first?

No, go ahead. And then I'll just follow up both you and Steven.

Okay, so you asked, how does the critical path, the linear pedagogical path that is traced by this version of this guide, reduce the actual or perceived impediments in an audience specific or in a general sense? So it definitely helps us see where math could or should or does or will enter the picture. I think it sets up an aboutness of the math. Like, what is perception about?

This is a graphical interpretation, graphical meaning visual, but also like a network, Bayesian graph. And so if you're concerned with a different connectedness of terms, this is not the math you're looking for. But on the perception side, if this is kind of how you're setting up the template for what you're wondering about, like what's out there? That's what this first image shows with W and X . What's out there? And how do I act or how does an agent act? That's the aboutness, which is not unique to active inference. That's control theory, cybernetics, etc.

The stepwise guide element, it reminds me, I hope it's not unrelated, in high school cross country, the coach would say something like, well, the race is going to start and you got to get out there fast. And then it's the middle is the part where you need to be running and going hard. And then at the end, you got to push it and finish and run hard and everything. And I was like, you could have just said run the whole time. Was it not the same thing as saying run at the beginning, middle and end?

And so, but run really hard during the whole race. I don't know, coach, a whole race is a long time to run.

[illegible]

[illegible]

Dean? Yeah, and I wouldn't disagree with anything you're saying there. But I do think that there's a not too subtle difference between stepwise and then the kind of step function, the leaving of the critical path, and now seeing a set of mathematical formalisms around a statistical distribution. And I'm just wondering how that transition occurs, because it doesn't seem like there's a natural ramp from following something clockwise or counterclockwise, to being able to change things into alpha alphanumeric ratios, which is essentially what the statistic when it's written out statistically, what that does, right? The sum and the ratio. So if the impediment was some of the non-frequent math, and I'm not critiquing in any way what Stephen's written out here in his guide, but it seems like there's still some assumption on the author's part, authors, plural part, that you will be able to take enough from this critical path representation to see what it looks like when you're overfitting. Like I like that there was a lot of explanation of what overfitting is, and then there was Plunk, and here's the formula that that overfitting looks like in mathematical form. So it's kind of a big assumption that the person who's reading this is going to be able to make that leap, go from this level of walking clockwise or counterclockwise, to now suddenly being able to understand what all of those different symbols in that formula means. I'm not saying it can't be done, but I'm just wondering how, as people who are trying to lay out something in a kind of step progression, how they assume that they can just sort of put holes into that big wall and people still don't have a fear of heights, right? Like it's still but it's still better than not having a ladder, but it doesn't mean I want to go up 40 feet without safety, right? Just because it's there. And I kind of see that in terms of how the transition works between this kind of this form of representation and then what the representation looks like in the math. That's a big leap,

Thanks, Dean. Steven?

because data kind of implies an inherent representational nature. And they talk about this in the paper a little bit,

of that tension between something that is coherent enough and can be represented, and something where you can't ever quite get at that. The inference of the predictive processing has to then leverage something like

generative models using the body, etc. to even get at the, that, that, that, that, that, to allow that arrow to be something in a much more fuzzy way, you know? And so there is always that risk of leading people astray, even at this early stage.

So a few comments there. Dean, you asked like, how do we get from this graphical layout? So we've taken the first

byte, which is, this is roughly the correct topology for thinking about problems of perceptual inference and action inference.

That is, if someone re-imagines this, amazing. But this is just the approach to perception, cognition, and action, and impact that is a common kernel of a phrasing.

Now, let's just think about it as an example, rather than just the abstraction. Let's just say that we're interested in

listening to a cat, and then inferring what its location is. So we're only talking about this simpler W and X connection

first. So here's what the empirical data are. 4, 2, 1, and 3. You could have a model of decision making on X that just says,

the cat is always in the kitchen. Just, I don't care. I'm okay with being wrong. The cat's in the kitchen.

Or someone could say, the cat's always in the bedroom. Or I flip a coin. Or I go with what

some other person told me. There's many ways to make a decision about world states conditioned on

data. One principled approach, which they write here. One famous method for solving this problem,

Bayesian conditionalization. So that's the question, which is framed normatively here, about what one

ought to do, is integrate their beliefs, which can be empirically grounded. That's parametric empirical base.

So, let's just say, the next step is, we can use Bayes' theorem to look at inference on external

states given incoming data. Again, if someone disagrees with using a Bayesian approach, an exact

Bayesian approach, that's the time to speak their piece. This is a very simple example,

example, with like kind of the minimum two by two matrix. If it was any thinner of a model on any

dimension, like if there was only a kitchen, then the question of location inference would be trivial.

It's in the kitchen. If there was only one sound, equally, it would be a trivial problem because the

sound would then be uninformative from an information theory perspective, which they're not bringing up here yet, about location. Then the next tiny step, and then Dean, is to go from the exact Bayesian approach,

the parametric Bayesian approach, to variational inference, which is, imagine that this was not

just two by two table, but there was a higher dimensional state space of observations. And so,

it's not going to be possible to just look at this in a spreadsheet or on a piece of paper, but it's

actually a large model. And so, we need tractable heuristics. And so, that is the next tiny step from

now we're going to take not just a principled approach to treating the w to x relationship,

Bayesian perceptual inference, but we're going to go beyond that and ask what's a tractable and reasonable approximation, variational inference. There's other situations where maybe there's another tractable approach, like MCMC, Markov chain Monte Carlo. That's another approach, which we've talked about in other cases. That's the pointillism sampling-based approach. It works really well for certain problems. It doesn't work as well for other problems. And so, that's the next little tiny step. We're going to go from this setup of perception to using Bayes' theorem. You could do otherwise. And then, from using exact or parametric empirical Bayes to using variational Bayesian approaches. And again, you could do otherwise. Dean and then Steven. Yeah. And again, all I'm, I think I'm pointing out is that I, for some, it is a tiny little step. But I don't think it's a ramp going from one version of representation to four square version of representation to formulaic version of representation. I think for people who see the math maybe as an impediment to their further continuation of using the potential of the free energy principle, maybe they aren't such small steps. There certainly aren't ramps in terms of how we interpret each of those three particular ways of representing information. There's no ramp. There's actual, an actual transition moving from, from what you've got on the screen now to the 60-40, 50-50 way of being able to interpret that way of representing the information to the one that has an equal sign. They're very, very different, right? Like, I mean, I understand the relativity piece and all the, and I, I see that carrying throughout. This is what relatively, relativity looks like this way, and then this way, and then this way. But I think, I think the struggle for people who are putting guides together is the assumption that what's little for you is little for me. It isn't always that way. Now, I'm not, I don't know if we can fix in that here in these labs, but I don't think it's something that we should assume is necessarily one, one small step for humankind. Because I think the reason why the authors were asked to put two years worth of work in this in the first place was because the math is acting as an impediment. Now, do you start with the math? Because I agreed with Stephen last week when he said, this is the direction we took. We decided to slow

things down, not simplify, but slow things down. So I really appreciate why they've done what they've done. But if they knew going in that they were going to have to go slow, then they're probably going to see whether or not that bump, as we change representations, is actually relatable to the previous version or form that the relationship was presented as. And I don't know that people necessarily do, like, Blue's going to have a say on this too, but she mentioned the arbitrariness of something that doesn't update. Right? And so it doesn't retain its form. And should we just assume that we all saw the morphology in the same way? It's tricky. I'm just going to leave it at tricky because I'm not sure anybody that's putting a guide together necessarily has the perfect way of being able to deliver it. Okay. Thanks, Blue. And then, Sue.

So something that I think is interesting is the mathematical interpretations have, like, philosophical underpinnings almost, or maybe it's the other way around. So when you're looking at, like, for example, this difference between a probability distribution and a preference distribution. Like, if the cat is in the kitchen 75% of the time,

you can just assume there's, like, the underlying assumption that it prefers to be in the kitchen most of the time. Right? Like, so because that's where the food is or whatever. So, and similarly, like, in homeostasis, if I'm at 37 degrees Celsius, that's my temperature. If I'm there 95% of the time, you can assume that I have a preference to have a body temperature of 37 degrees Celsius. I think that there's that underlying connection. But it gets really complicated when you try to extrapolate to something like cognition. There's, like, I prefer to be rich. I prefer to have 90% of the money in the world. For example, like, you can say, like, I want all the money. Like, like, you think about something like that. But then there are so many complicated, like, do I really want all the money? No, because then everyone else would be homeless and impoverished to be horrible, right? Like, so you don't really want all the money. Like, there's, and then if I say, if I prefer to be rich, but I'm stuck in an impoverished situation, you can assume that there's, like, a learning gap between how do I earn money and being in the space where I don't have any money. Like, I don't know how to have money. Like, there's a knowledge level. So when you, when you bring cognition into it, and like, you ask people, like, what their preferences are, like, I prefer to live by the beach, but I live in New Mexico. I mean, we have lots of beach, but no water. So there's like, you know, there's, there's, I don't know, there's the preferences and then the probability, but, but is there some kind of like cognitive dissonance? Like, you think we think what our preferences are, or maybe not where our probability distributions are? Like, I prefer to be by the beach, but 90% of the time I've heard of you by the ocean, but 90% of the time you find me in the desert, that there's like, there's some cognitive dissonance going on there. And so I think that the mathematical interpretations have a lot of philosophical implications that maybe, you know, we need to think about more, like, when is a preference distribution and a probability difference distribution different? And why are they different? Like, what is going on when the preferences are not the probabilities?

Thanks, Blue. Yes, this will take us soon to the tale of three piece. Steven, first.

David, first.

Yeah, very good points there. When you said, Daniel, there about the exact bays on the world, it goes into that. So it goes into the variational bays, which in some ways, as it starts to hit the Markov blanket and start to give time steps as it starts to go through into the generative model, you then got, and as was useful from our, again, the symposium with Carl Christen, he talked about, you know, the starting point often for active inference is the generative model, you know, how do you get at the generative model? So the general route is that in some ways, this exact bays is almost needed in the modeling sense to give you what we think we know the world is when we're setting up our model or our experiment of people doing something where we're going to see what the results are. So it's like the agent can't know what W is. They can't know what the exact bays is, but like with the cat experiment, sorry, the cat matrix there, okay, we could arbitrarily put one in that we kind of can say we know as objective observers. It's going through, goes to the agent, and then the agent does something. Okay. And, and as blue said, sometimes that,

that what they do isn't only in response to what's coming in, which is quite often not, it's not just, can I, I've seen food, can I eat food as much food as possible? There's this, well, what's the consequence of my actions going forward? Maybe a lot of times we then start to get into this higher dimensional stuff where the circle that you put in there going into W, there's that red circle on the left, which sort of goes into the W is what is really of concern. You know, the humanities around what's, what's that about, right? And a lot, normally the generative model work that is done in active inference is, is the circle on the right hand side. I, how do we predict the way beliefs are being updated and what's the way the beliefs, because it's normally coming in that direction because you can sort of control, you can do an exact bays. There's the one place you could theoretically set your experiment with an exact bays is W. It's kind of hard to put exact bays on an agent, unless of course it's a robot potentially. So I think there's something quite interesting there about how much is this, and it's a practical instrumental tool, right? How much is this about the way that modeling happens and how much and the maths to make that happen and the philosophies make that and how much then is this broader and it may be that it gets recapitulated depending on which parts we simplify or allow to be more complex or want to get a purchase on out of those circles. And maybe you just explain why you put the circles where you put them just to see if I'm getting your, your rationale, but I think that was kind of useful to put them in. I'm just highlighting the directedness of these statistical dependents. This circle of action policy influencing unobserved states, we've seen it in the partially observable Markov decision process as where? That's B. W changing through time is B. That's how hidden states change your time. State one, state at time one, state at time two. Then Z influences here in this simple only four nodes. So if you only get four nodes, you draw the dependency that way. Like my action to change the thermostat in the future will change the temperature. When we can unpack it a little bit more and move it from this sort of minimal cybernetics or control framing into some of the little bit more built out models we've looked at, that is where policy selection influences B. And so yes, there's enormous concern over how our actions influence things that we can't directly observe, save for by proxy information incoming back to us. And that is not a unique problem to active inference. Dean, Evan and Steven.

Dean, Evan, I think I'm going to circle back to what Blue was saying because I think she's hit on something really critical.

Dean, Evan, I think the whenness of this, I think is what I won't say was the was the thing that Carl started on when he took his first nascent steps into trying to put something that we can we can hold on to, something that we can look at philosophically and mathematically. I don't know if he was thinking about whenness at that point, when he started down that path. But I think the more we talk about it, the more we realize that

as a for example, in a guide use, I think a person who reads Stephen and the other author's work have two questions. The first one is my use of time, the number of minutes I have to commit to reading the guide before I can then say whether or not certain impediments have been removed.

Well, at the same time, and this is what I think active inference does, it also allows us to invert that

first accounting of what we are doing with our time, invert it from use of time to time of use. Like, so if I do commit to reading this guide, what's the shelf life of that? Like, how long am I going to be able to take the my takeaways and actually be able to apply them to different concepts? And then there's a fleshing it out part, which I think Stephen's kind of talking to, which is, so when we take all of this information and we embody it, what does that feel like? What does that, what does that read out in terms of our behaviors, etc., our reactions, our responses? So I think ultimately, if we're looking at the whenness of this, we have to start talking about heuristics. And Stephen specifically spoke about that. He said, I wasn't, I think I'm paraphrasing him, but I think he said, we were concerned about whether starting with simple doesn't also have a cost. We know that it has a benefit, because it doesn't overwhelm people. But if we are going to look at simple rules like heuristics, does that also have a cost? Now, he didn't answer that. I was just rewatching the 37-1 before we came on today. But I think that's something here that Blue has now touched on, and I think Stephen is touching on as well. If we're going to put a guide together, when did the, when did the rules fall out? Do we assume that they fall out just by picking the guide up? Or do they, are they something that, that the reader or the user manufactures after having the experience? That whenness, I think, is what this active inference stuff is touching on. And, and, and, and it's, in some cases, it feels pretty ruddy arbitrary. It's hard to tell the when of the heuristic. Do I start with the rule? Is my prior, my priority? Or am I really trying to update? And I'm trying to come up with something that really applies at some point in the future that I haven't arrived at yet. And I know that sounds kind of not answering anything. But even, even with the statistics, it's hard to figure out what each agent is going to do, depending upon what their biases. So they want to start with the rule, or they want to develop one as they go along. And as the *Bacillus axillus* in 34, *axillus* bacterium, showed, exact base doesn't mean you get it exactly right. It means you're following a process, and with maladaptive priors, even accurate information coming in can result in a maladaptive decision making.

So this is not Panglossian or Pollyannish. Just because there's an approach doesn't mean that it succeeds.

Perfect. Right.

Yeah. So Steven, then Blo.

Yeah. So two points that sort of speak to that. Well, I'll say the second one first, because it was in mind is when we are, when we see the modeling, we often see, say, we talk about the *Bacillus*, or we see the way that modeling happens. High order agents, such as humans, tend to find that the W is fixed as a, like the cat is either purring, or it's meowing, it's in the kitchen, it's an exact, or it's in the bedroom. Right. So it's a kind of defined, exact base kind of context. And then that allows the the other parts to kind of still find some very interesting and tractable, useful trends. Right.

So as it goes through now, when it goes, then the other way is, okay, how do you get into something which isn't exact base? And you tend to see it turn into the swarming of, of voxels, the small, you know, Lorenzo tractors or *bacillus*. So very, very, very simple, agental

scenarios with very little degrees of freedom, because, and this is in a way trying to address that challenge of, well, you know, what if you don't know if it's in the kitchen or the bedroom, it's in the doorway and it's, it's wandering around and there's the time, the timing between getting sensory information and my, the action is, you know, isn't in step. There's a lot more bleed out, you know, and that becomes this kind of, how much is the temporal, I saw this word from, from Casper Heads, leakage. Because there's a leakage out of that sensory, sensory, I don't want to use the word input, but the statistical dependencies coming in and the statistical dependencies going out, there's a, there's a leakage going on, which makes the modelling kind of challenging, though. At some point it becomes, okay, well, it's overwhelming, right? The, the, we might philosophically say, somehow we're able to do something like that and, and our biology has enabled us to overcome that challenge, but to actually put it into a, a model like the one on the right, top right there, it starts to become, and in a way maybe that's where effect comes in, as maybe, okay, that, that may be also true for us, that's why we need something like affect, to hold together this sort of fuzzier kind of knowing, which can't be resolved in that such a solid way. So that's the first piece. The second piece, very quickly, is you put the, you know, the, these, that, the, the red circle that you put next to the X there. So in some ways, that, that is what's, like, while it's sort of, we're inferring W, what we're really doing is we're inferring, we're either inferring the circle there that comes in to the X, or maybe, and let's assuming what the arrow that's going into the agent is, is what's coming out of X, or is coming from X. So that's in a way, the edge, isn't it? That's the kind of boundary that's being, the beliefs, when you say B, the B is about that, not strictly about W.

There's a belief distribution on W and X. W is latent, modeled, unobserved states, and X is modeled, empirical, observed states.

Right.

Okay, Blue?

So just to kind of riff off of what both of you guys were saying, maybe this is tangential, but, but when you talked about leakage, Steven, it reminds me of like the information partitioning that we discussed with Majid Beny and how there kind of always has to be some kind of leakage. Like, if you have a complete information partitioning, you're actually like, in a, in a vacuum or a bubble, like you're, you're existing, there has to be some kind of leakage across the partition all the time. Like, and I've, I've come to realize this, like, we have to be sharing information with people that are, you know, how do we form a Markov blanket? How do the four of us come together and form this live stream today? Right? Like we're sharing information, even though there's a Markov blanket that exists between us. And for some of us, it's thicker and thinner. Like Daniel and I will be often like doing the same thing at the same time, like, like even like, you know, separately, just because we, we work together so much, but there's, there's leakage is there. And then it also made me think about the maladaptive behavior that Daniel was mentioning. And so when you have this, like, we always wonder what is the cause of this maladaptive behavior? And this is something that we were discussing maybe a couple

weeks ago with Jason, you know, that, what is the explanation for maladaptive behavior? And it always makes me think about the scale, right? Like in, in the situations that are utilitarian, like, you know, maybe someone sacrifices themselves for, you know, they jumped in front of a train to stop the train to save the 300 people that are on the train. And so there's the maladaptive decisions do occur, but, but perhaps it's at like a societal scale or, you know, like in a body, you know, one piece will die off to save another piece. Like even in the brain, like the formation of like the glial scar, I think of that, like, which it's like, essentially like, an, it, it puts a hole in your brain, but it's to protect the rest of the brain from the cell death that's happening in that, in that center hole. And so, you know, it's, it's, there, there is this information leakage and, and, and even maladaptive behavior that I think are, are, are scale dependent, scale friendly.

Ace blue, um, Dean, and then we'll move to another section.

Cool.

Yeah. I, I don't know if we're at an inflection point with this, but I, I do want to bring this up again. I, I think that the leakage speaks to another metaphor that I've brought up way too many times last week, which is, I, and the, in the point zero, which I think there is a filtering piece to this. I mean, this morning I put the raspberries in the sieve and I assumed the raspberries would stay and the water would pass through. And there were parts that I didn't need, like I wasn't trying to make raspberry water. I just wanted to eat the raspberries. And I think that if we understand that there's two, a minimum of two aspects to when we're talking about active inference, I think we do ourselves a favor. And I think we do ourselves a favor for writing guidebooks about active inference and free energy. Okay. So I think, I think the guidebook for the guidebook, if there was, if there were, if there were some pillars that we could put a guidebook on, here would be my suggestions. First of all, there is going to be, because we have a Markov blanket, we have a separation, that differentiation and that integration are going to be inherent in however we observe something and then respond to it. Second thing that I would suggest is there's a certain blind spot removal piece to this. And I mean that in the, in the context of say something like change blindness, because of over attending or under attending and where he talked about over fitting or failing to explain. That change blindness, now becoming a greater awareness availability, because we don't necessarily have to lock in prematurely or wait too long. There's kind of a sweet spot that active inference is, is trying to say exists, not perfect, but it's, it's better than not having a sweet spot. Third thing I would say is if we're ignoring the foraging aspect of this, if we expect that the framework is going to be another frozen meal, then you're going to get the quality of a frozen meal. If you're willing to forge around the perimeter of the store, you're going to come away with a whole different experience. It's more time consuming, but in the end it's like learning to fish. It's, it's, it's a whole different way of trying to process those uncertainties. Right? So I think at the end of the day, what pops out or when, when we decide that we can actually say somebody is using active inference, whereas before they weren't, I think

what they do is they come up with a new sense of what rules, which are being updated, which are being modified, which are being iterated, and sometimes even being extrapolated, we have to really focus on what that means under particular circumstances, as we move from place to place. So again, I'm not sure, maybe that's, maybe that's not being fair to the people who wrote a 60 page guide for philosophers, but I think if we're going to look at the, the foundational things, they're going to be able to tell whether or not we're actually doing and using active inference, I would think that at minimum, those four things are going to be present. Starting with the filtering thing, because I, again, if it's not a minimum of two, I think we over reduce, we put a frame around it, but we don't actually know what's actually a gradient descent and dynamically stable within that. So just something to think about, if we're going to write guides.

Thanks, Dean. Okay, Steven, specifically on that, and then we'll go on.

Yeah. Just referring to this idea of screens, and it comes back to the idea of screening off or filtering, or I think that the X and the Z here, there's a question we often, in this particular context, they are talking a lot about screening it off. And it seems to be, whereas in other contexts, say, Axel Constance work, I feel like it's trying to extract as much as possible from the informational potential in an environment. And I think that's important, isn't it? There's a very different, I've worked, I actually used to work as a print broker. So I actually fire up these big presses, and it takes a thousand copies to get the magazine actually starting to print properly. Like the rollers and the blanket has to get inked up and everything. And you've got all this waste. Eventually, you actually get something that looks good. And now it's running. So you've got a thousand wasted copies, basically. You're trying to extract as much as possible from the inked up print head, print cylinder, right? This is on a big press. Now, in that case, then you're trying to get as much information, basically, across into the paper. Now, there's other cases that you can imagine where it's, I don't know, someone's taking a video in a, and there's bright lights going on. And there's, you know, you're trying to filter out, like we say, you've got bright light, you know, you're looking at the sun, you've got to filter out to see anything. In that case, you're still trying to get your sensory data in, you're still trying to work on the action policies. But both X and Z in a traditional blanket is a dead thing, you know, but how much is X and Z? You know, how much is that a living kind of process at certain times that is moving? I mean, you could say it's doing a bit of everything, but how much is it trying to screen out noise and bringing, and how much is it trying to extract, you know, from Shannon entropy and stuff like that? So, you know, I thought I mentioned that. Yeah, it makes me think of like, being on the phone with somebody, and it's like, you're both trying to pull as much information out of that limited channel, put as much into a limited channel. And so let's go on to justificatory links between or among dialectical categories. So we'll do some slide play. We have three types of claims. Dave, I hope I'm using the word appropriately that they're domains of claims. Mathematical, red, empirical, blue, general, green. Okay, so no commitments, but can somebody just like give a specific, concise example of where we see one of these cross category statements? It can be in any direction amongst the three. Where's a case where somebody uses a mathematical, the first part of

their sentence is mathematical, and then the second part is empirical, or vice versa. Where's a situation where the first part of someone's sentence is mathematical, and the second part is general? So, I'm sure that many have been said even today, but let's try to get a few really specific examples so that people can just learn to identify these kinds of sentences, and then we'll be evaluating what they mean. So blue, go for it. So nothing that's been mentioned today, but you know, the p-value of 0.05, you know, demonstrates that whatever my claim is in my experiment is significant, right? So that's something that all the time you see, the statistical, you know, justifying the empirical, and less so. I think we're coming away from the significance of the p-value in scientific literature, but you see that a lot. So where would we put that kind of, where would we put statistics? What domains does statistics jump them up? So statistics to me is math, like statistics is the sentence that bridges, that's the field that bridges mathematics and empirical claims. Like, so when you make the sentence, my p-value of, you know, 0.05, $p < 0.05$ demonstrates that whatever my claim is, my empirical claim is significant. So for me statistics is that bridge between math and empirical claims. Awesome. Yes, totally agreed. We looked at these two categories of ant, and we saw this one doing it six times. This one did it two times. We did some math on the numbers six and two. You can't do math on the ant. You do math on the numbers six and two. And from that mathematical claim, we're going to say something about those two categories of ants. So yes, mathematics going to empirical is a lot like statistics. Okay. What about empirical going back to mathematics? What is a specific natural language expression where somebody begins

where the warrant of the rhetoric is empirically driven, meaning based upon observation, and then the consequence of the argument is mathematical? How about because the organism is maintaining body temperature

homeostasis, empirical, it is the case that it is doing predictive processing or active inference?

So here,

um, it's now there's so many ways to even enter here, but this would be like from results.

That's the hallmark of starting in empirical is you're starting with something that is almost either trivial or just simply a fact or observation about the world. So from results to what?

Um, some sort of statement about... Now, what's interesting about the way that the authors did it here is they have no... None of the three cases have mathematics in the second position.

And that's kind of like the XKCD comic where mathematics is like, oh, I'm way over here.

I can't hear you. The fields arranged by purity meme. Because mathematics is almost framed as like internally coherent. And then information comes out of mathematics. But because it's theoretical or abstract, there is an internal logic in mathematics called proof. And then there's very little need to have something enter mathematics, which is why it's a... We'll see as we start to flesh out this, um, shape.

Stephen?

Yeah, as you're doing that, I'm curious what you think about something like gambling.

Say someone is, um, you know, betting on horses. It's the statistics of the horse gambling.

And then there's their motivation for gambling. So in terms of the empirical, you can see, okay, the behaviour

is fairly consistent, certainly over a type of gambler at horse race. The math can be established.

But then there's, I don't know whether it goes into the general, well, what's people's motivation for the teleology? But maybe I'm jumping the gun there. But am I on the right lines there?

In terms of how would you put that into this, um, scenario?

Can you restate it with an antecedent and a postcedent? Just a specific rhetorical nucleus?

Okay, so there's, um, there's a group of people...

Because X, then Y.

Oh, because, because, um, there is a group of people at a horse racing track, um, and

Because we know that these horses have performed in a certain way over previous races, we can predict the probability that the horse will win or be told to people to win as being X. The motivation for people to make that inference about and the behavioural choices is maybe more general.

But that seems...

Yes.

We'll get to the general edge because it's very connected to math. Okay, Dean?

No, I was just going to say, I like the word results there, but I,

I think what, what it's implied is, is that there were a certain number of observations taken that then moved the formalism to something specific? Is that...

Am I just confirming something or am I putting more in there than needs to be?

I had a really similar question. Um, like, so, um, when you're talking about going from empirical to math, to me, like, you brought up, um, the, the idea of, like, a probability density, or, like, a non-equilibrium steady state, and, like, that's immediately what came to my mind. So I don't know if, if that's, echoes what Dean just said.

Yeah, it does.

Let's get at the distinction between general claims and mathematical claims.

So, math is sometimes seen as general, and when ideas are framed mathematically, indeed, there's a lot of overlap, but the authors do separate it out. It's in section 4.2.2.

So, what would be, like, an example of going from a mathematical to a general, because math, then general, or because general claim, then mathematics?

Well, you can, you can talk about the rate of, the rate of, the rate of acceleration due to gravity, and then, then make the general claim that, that, uh, if I, if I go from, from this height to this height, I can, I can expect to get the ground at a certain rate of speed or something, I would imagine.

Okay, great. I'm going to also pull out just two examples from the paper.

Yeah, from the paper.

So, here we go. So, we're in 4.2.2. So, there's going to be two examples given.

So, again, we're looking here at math going to general, and then we'll see if we can run it back.

So, how can mathematical claims justify general claims?

Ramsted et al., 2018, assert,

The FEP is a mathematical formulation that explains, from first principles,

the characteristics of biological systems that are able to resist decay and persist over time.

First part of the sentence, mathematical.

Second part, general.

It rests on the idea that all biological systems instantiate a hierarchy,
a hierarchical generative model of the world that implicitly minimizes its entropy by minimizing free energy.

I almost see the reflection in that sentence that because of the hierarchical generative model,
which is a general claim, though a formal one, then there is a mathematical implication.

So, maybe we can put this Ramsted 2018 here.

Okay? And then here's the second or a second example that the authors raise in the paper.

Axel Constance's contribution to this topical collection uses a mathematical claim to make a general claim.

On the basis of a numerical example, he argues against the claim that minimizing free energy entails life.

Rather, he believes the converse is true. Life entails minimizing free energy.

We read and loved that paper.

And that's very much related to the, quote, like, one ugly fact destroys a beautiful theory or something like that.

Now, is an example mathematical or empirical?

If it were just on the basis of a family of equations, he argues against the general claim.

But then again, isn't even the family of equations an empirical observation from the domain of math?

But we'll put it in the section that they had it.

So, in these cases, we're using specific math examples.

And that was sort of the provocation of 34 and Axel's paper, which was like, I did a math example.

So, that's kind of resting on both the empirical and the mathematical.

I did a math example, and now I'm going to make a claim about a general topic, which is the relationship between free energy minimization and life.

So, that is certainly moving in this direction.

Whereas, so, because math can be abstracted and generalized, there's often similarity here.

Because math rests upon specific examples, even if the examples are, some examples are more general than other examples, it still can be said that they're empirical.

Not exactly in the way that cat location measurement is, but still, it is the case that math produces results.

There can be a results section in a math paper.

What about this third edge?

What's the relationship between general and empirical?

And let's leave math out of it.

Blue?

So, this is something like, I think about, like, what Dean was saying.

Because acceleration due to gravity is, you know, negative 9.8 meters per second squared.

When I drop this apple, it will fall to the ground.

Like, that goes maybe empirical to general.

When I drop the apple.

Because when I drop the apple, it falls to the ground.

Comma.

Gravity exists in the whole universe.

That was quite a jump.

Well, because apples fell in.

Not down.

That's flat earth.

Because apples fell in.

Gravity is everywhere.

Okay.

That's one example.

So, then I was thinking of, like, a sort of ants evolution example.

So, the general to empirical would be, like, because of the way that evolution by natural selection is, ants optimize task efficiency.

And then, the other direction would be, like, because in my experiment, 30% of the time ants were observed to be doing this, selection has been acting on ants.

So, these are very interesting because, are there some edges that we rest upon more?

Like, what if we could just number these edges, 1, 2, 3, 4, 5, 6.

And then, annotate.

People could have different perspectives on it, but we could annotate papers in different areas.

And, okay, if you're just doing 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, it's like a one-note paper from a structural rhetorical perspective.

Under what situations is that preferable?

Not preferable?

Are there certain edges that are weak?

Are there pairs of edges, like a chord, that we want to play together?

Just like the Ramstead, it was, like, sort of forwards and backwards.

Law of the jungle.

Law of the pack.

From the mathematics to the general.

And then, instantly, a grammatical reflection from the empirical into the mathematical.

Okay, Stephen, then Dean.

One question.

First, I just asked, empirical could be seen as exact base.

Empirical means observed.

Observed.

So, the exact base, if you're just taking however you lay that out, if it hasn't been computed beyond observed.

Okay, so effectively that's true.

Okay, that's the first thing.

And then, general, in most senses, is what we might call inflated or higher dimensional.

So, we go, generally speaking, the models are empirical from observed observations into math, into generative model, into some general pattern of action selection.

However, it could be possible, but generally it's intractable for complex situations to go from general to empirical and then try and get the math out.

Which is kind of what Alex is saying.

It's like, there's a general principle that empirically plays out, but good luck finding the math to show it outside of a bacteria.

Okay, Dean.

So, I find it interesting, Daniel, that you, just to keep the conversation sort of alive, is you talked about the edges of those three circles.

And I think that the task of being able to find examples of those bidirectionalities, again, goes back to that question of, so, what are the rules?

And how fluid are they?

And as we're moving from one node, so there's a change, the moving from node to node, we're building that relationship, we're building out these rules.

Is that one of the great byproducts of being able to maneuver and manipulate, not getting stuck with only a sense of unidirectionality or oneness?

Does this reinforce the idea that if we're going to actually really be able to demonstrate the use of active inference, for example, as an instrument or a tool to be able to gain a greater awareness?

Are we really then talking about how we not just find examples of rules, but actually form them?

Because that's what you did, that was the exercise that you just went, you just built a whole bunch of rules.

I know you, I know we've laid them down on the slide as examples, but that those relationships now are, they're not even bridged, right?

They've now got some sort of a pairing effect.

So is that, is that, is that, is that, I'm looking today back to the, and then what happens, what I'm looking today for is, so we guide our way, we guide our way to what?

Is there, is there now an awareness, a sense that we didn't have before we went through that, that effort?

And is, and is the thing that kicks out rules or is it just examples?

What do you think?

I'm asking you, Daniel, because you, you did the most of this work.

Do you see the rules or do we just see the examples?

I just think about how in a nested cognitive model, the outcome of one layer is like the inflow of another layer.

So even a rule can be a rule in one layer or application, but yet it's also an example of a rule.

Mathematics and general claims tend to be more rule-like, but again, even rules are examples.

Whereas empirical, it's always the exception that proves the rule.

Like it's the exception is the observation, which transcends rules because if it happened, it had to have

occurred.

And so the empirical to me is like examples when you're in this domain and then movement away from the examples in the scientific empirical sense pushes us into the rule world.

So here, this red line is going to be more example-like.

And so in that sense, like with the caveat, again, that even rules and generalizations are examples of.

But in the scientific empirical sense, this is more like an example.

Can you hear in your own explanation the sort of just a factory link between example and rule?

Do you hear yourself justifying the difference between how we might look at math in general as being different than the empirical?

That's all I'm trying to bring out.

And again, it's a not, it's not, it's not, it's, there's nothing, I mean, it makes perfect sense.

But I think if we're going to find, if we're going to find that somebody used a guide and what is the, what is the, what is the result of that?

It's going to be that people have a better sense of what the rules are and what the examples are and what the relationship is between those two things.

That would be, goodness gracious.

I mean, if I'm talking to anybody between the ages of 22 and 62, if we could get there, holy mackerel.

Thanks, Stephen.

In terms of, in terms of opening up our understanding of things, right?

One thing I think with what you're saying there, Dean, is, which is, because it does become interesting.

And I, uh, something Daniel mentioned when we have multi-scale, the actual nature of the rule example actually can shift.

And as teleologies shift that, that I think is kind of revolutionary in a way when we start to think about, there's a variation in the nature of what rules are present, you know?

So, you know, the sort of rules around what it means for me to do something in my environment, um, my cells individually, I can't play that game.

My organs can't play that game individually, right?

So, um, there's these, there's, there is, um, this, this challenge, right, of integrating, or maybe this is idea of, okay, well, as well as saying what's happening,

um, what, where is something or what temporal and temporal and, um, temporal and spatial scale does it hold?

Right.

And I think that may be something that's kind of interesting is where, so generally speaking, people go to the racetrack and, you know, they behave in certain behaviors at the scale of, um, their, their, their, their part in society.

But they, um, and we have empirical data to support that.

However, um, at the biological level, um, it may be something to do with, um, adrenaline rush you could talk about, or you could talk about, um, the fact that they just happen to have a nice sandwich bar and someone enjoys getting the food, right?

And they don't even like gambling that much, um, whatever it is, or wearing a funny hat to ask off.

So there's, there's, there's something there that could, how do we, I wonder that, that, that multi-scale question, um, cause I'm very caught with the idea that you've got Axel Constant taking very, very small.

We have, we have this incredibly small world being addressed with active inference.

And then we have quite sort of, you could say gross examples at scale, which ends up revealing a lot, you know, at larger scale.

Um, but, uh, I wonder what you think Daniel about how, what, what happens.

And that, that reflects what Daniel, I think was saying about nested systems and they're, they're sort of informing each other in a way.

Like if I, if I think I'm going to win some money on the races and I better get myself sorted, I might sustain my stomach a bit more.

So I've got a better glucose level to make a good choice.

Right.

So suddenly there's two different levels coming into play or just different scales.

So, um, I don't know if that's, I don't think that confusing things, but I think it does make the modeling question because active inference is ultimately a modeling process, um, uh, which is trying to get at, I would say it's trying to either get at the general or to, to,

predict an empirical.

Um, if I, if I'm looking at this diagram, but maybe I'm, I'm mistaken there.

Right.

Where is active inference in this?

Is this the trace of active inference?

Is this the map?

But then we play on this territory.

what is the adjacent possible?

And so we can ask first, where's active inference here?

And when is it here?

And then what next?

Dean?

Yeah.

This is where I think it's, it's about, it's in the relationships and how we decide to parse those relationships.

And it seems rather arbitrary.

And I think it's one of the reasons why we're, we can always disagree on when the rules are in effect and when they're not in effect.

I mean, it's happening up here in Canada.

We've got all these rules in place and then people just decide to pull up big trucks to the border and go, no,

we don't like your rule.

Right.

And so now there's being Canadian, now we have to figure out how we're going to deal with that because we don't have any guns.

So I guess we're going to have to disassemble all the trucks or something by hand.

But the thing is, is that if you actually look at this active inference is in the relationship.

And I think it's, and then, and I think to Steven's point, if we, if we think that a rule applies without limits, that's the scale free part of it.

There, from a mathematical standpoint, we can, we can find cases where that's true.

But then there's a very scale friendly part of it, which says, I don't let this rule doesn't seem to make sense in this specific situation.

And that's where Axel came forward.

He, he essentially said, I'm sorry, but, and, and so did, I'm sorry, the, there was another author we had very recently who, who, who agreed and said, there's, it was Majid.

Majid said, look, there's, there's a piece to this that says there's something that's local and proximal and specific.

And why are we pretending like that, that fits the same context as scale free?

So, again, I think I don't see that means that active inference is useless.

I think what it means is it can tell, it tells when we can differentiate and when we can integrate.

Okay.

That's relational.

So one thing this made me think of is that reading a paper on free energy, free energy principle, active inference, look at the roadmap.

There's no dialectic within a domain.

The claims are listed, but it goes immediately to the relationship.

So it's like minimum two domains.

Okay.

Now it makes me think about the structure of scientific communication.

So in the results section of the paper, well, so the introduction might have a lot of general and empirical claims.

Selection acts on ants and look at their mandibles.

And we know this about niche construction and look at how they do it.

So that is like a very introductory piece.

There's not often formalisms in the introduction of the paper.

So that sort of blue arrow, general and empirical is very introductory in its scope.

Then the results section of the paper is like blue mentioned.

It's about like empirical and math.

We saw it do it 11 times and here's the p value.

And then we did this and we calculated the model.

And so it's like it shouldn't generalize in some senses.

And then in the discussion, there's probably all edges in play, but especially math to general.

Because it's now been generalized beyond the mere empirical.

Because we saw a p value in the difference between the nest mates of this type and that type, it does say something about evolution.

Or because there's something happening at an evolutionary timescale, it's consistent or inconsistent with our mathematical summary statistics.

Yes, that came from empirical.

But that edge is more in play.

Dean?

Dean?

And then in simple terms, that's how and when we fit PHIT, the rule.

And that almost, it does return us to the third P.

But I think it's, so we'll kind of close this section on the 2.4, whatever it was with the links amongst the dialectical categories.

Active inference as a transdiscipline.

All of these are often in play.

And it's almost like if the paper would have just been lists of empirical observations, there'd be nothing contentious because it wouldn't make any general claims.

That would just be called a data set.

If it were just general claims, it would be just philosophy.

And that's not to say philosophy is easier or simpler.

It's like it doesn't trigger anyone's domain crossing alarm.

But like mathematical biology, that does raise people's alarms.

Or generalizing from math, you're saying that because of the free energy minimizing, that there's a general imperative for information foraging?

Or relationship between the general and the empirical?

Of course, always fraught in empirical science.

Okay, you measured in that laboratory study with those undergrads and that stimuli, and you're going to say something about aggression?

So it's just very interesting because active inference does lay claim, or at least lay play, to all these areas that it raises alarms on all sides.

But how do we take that alarm pheromone and attention and then just align it with something that's going to be a public good rather than just let it fizzle?

Stephen?

That's very helpful.

And the nature of action and the idea of teleology, as soon as that comes in, I don't know if you can avoid being in the general.

Now, maybe active vision that's just doing a thousand-dimensional interpretation of a data set, you know, can

stick with empirical math.

You know, it's either recognizing a photo.

It's not recognizing a photo doing some deep, deep, deep computational processes.

But action, and particularly then action policy and then teleology.

And would you say that?

Would you say there's general claims or where would teleology fit in here?

Teleology is never directly measured.

Is that fair to say?

So it can be in green, but it cannot be blue.

One can say, I observed.

Sorry, blue.

But I chose the colors for a reason, too.

One could observe the animal trying to swim to the surface.

And you can say, well, the teleology is obviously not to drown or something like that.

But that's actually structured as because I empirically observed the swimming behavior, I am going to make a claim about the teleology.

So I would say teleology is in a general case.

Even another spin on this one would be empirical is like X.

This is the data that we're actually observing.

General is unobserved.

You can't observe things that are general.

You're only observing what Blake calls the minute particulars.

What is math?

Math is in the examples that we looked at.

It's like the process.

Oh, wait.

Active inference.

It's a mathematical process theory.

So math is a process of inference that, yes, connects to generalities.

And this is not hard and fast.

I hope it's not being interpreted in the most rigorous way.

But it does generalize because it has to be like the intermediating blanket.

So let's pull this W and X back in here.

All right.

So here we have the empirical states here being observed.

That's our data.

Okay.

And then here's the things that we would like to generalize on.

And then this gets to Dean's question about how do you go from this to equations.

And here's math as our sort of blanket in the middle.

That's when we actually do, whether it's variational base or whether it's exact or whether there's some other mathematical approach.

It's how we're going to go from these generalized or hypothesized things that we didn't observe.

How are we going to connect that domain?

And so it's almost like a little boomerang.

So there's probably a lot more to explore here too, Dean.

I just like how this has come together because math, as we said before, as I mentioned before, I think in the author's mind, the motivation moving forward on the paper was math could be a vehicle.

And it can move the process along because it is a process or it can be an impediment.

And the other thing that's really interesting here is because it's in the middle, isn't math got a lot of rules in it?

I mean, come on.

Right?

So, again, I don't think we're contradicting ourselves.

I just think we're asking, so again, what do we allow to pass through the sieve and what do we hold back?

And then what assumptions do we make about the sophistication?

What's the minimum amount of sophistication around the math process that one must have?

Dr. Carl Friston said, you must have a good grounded sense of what the math is if you're going to go forward.

Like he didn't couch his words.

He said, if you want this to become a vehicle and not an impediment, you're going to have to get familiar with the math.

So.

And sometimes, oh, the math is not the territory.

You know, important contribution and memes for sure.

However, if the debate were about whether territories were maps, it would be the silver bullet to simply claim that the math is not the territory.

But it's like, right.

And we want the map so that we can take action and be on our road trip or whatever.

So it's the math is not the territory.

While it's the conclusion or the second part of the rhetoric.

For some, especially philosophers who might be interested in what is really out there.

Because the math is not the territory, comma, then what?

That's like the dot two of act inf.

Which is, instead of just debating, well, yeah, well, because of homeostasis, what can you really say about teleology?

And because of allostasis, does that contribute to a new understanding of niche construction?

They're all important questions.

But then where do we go when we put the math is not the territory as our antecedent rather than our

conclusion rhetorically?

Stephen?

Yeah.

And like when you put here, so first the math, I totally agree.

I was thinking that a vehicle.

I was thinking it also takes a different.

So, for instance, say I have a million dollars.

It gets picked up, it's counted, it's picked up in a car, taken somewhere, and then it's counted as a million dollars.

The bit in between, it's not math.

It's just stuff, right?

But in this process theory, it's the, in a way, in a way, the math is enabling, and it could be an entropic process.

You know, it could be math in these, maybe some sort of virtual numbers way or noise way.

But the math is the vehicle for the data, right?

There is no data.

There is no banknote.

There is only a process which enables the general to the empirical.

And the empirical in this context is a modeled empirical.

In a way, what we're saying is we're going to take most of the models that we see.

There's a very simple general idea about behavior, say Ryan Smith's idea about how the heart behaves, how the gut behaves, something about how access consciousness or access awareness is achieved.

Some general thing.

And then we basically do what we might call a modeled empirical.

But you could flip it around.

You could flip it around the other way.

And you could have, okay, well, the real world, we could have high dimensional empirical outside there.

What's the general W?

What's the general W?

What's the general X internally?

What's the general, which is what we use as our heuristics, you know, which could be kind of interesting as well, which is maybe more in the action?

Because this is coming in on the sensory, right?

But I think this is quite helpful.

Would you say that's true?

It's a modeled empirical, what we get on the, in terms of how we work with it in the active inference?

I mean, looking back on it, is there anyone who disagrees that empirical is always as modeled?

Like, it's kind of funny.

Like, now I want to put a little red guy always intervening on this edge.

Little blue guy always talking about specific examples of math and generality.

And then, you know, the generalization is we use the, you know, in equal variance t-test.

So we generalized to connect the empirical and the math.

Dean?

I really didn't want to go there here, but I have to.

I was way too excited last week when I had a little thought bubble and said, wouldn't it be cool if Michael, or sorry, if Stephen Mann and Connor Hines, if we could get them in a room and instead of them talking about what they know, get them to go through this process?

Because that's essentially what we want to do is, is we want to have that little guy pop out as opposed to the one who knows so much in a dense way about their math or so much in a dense way about their philosophical guide.

Because that's when the true grasp of what potential is.

And it's not about always having the right answer.

It's coming up with the best answer in that moment, right?

And if we could get more of those kind of situations, I think people would actually, I think people will actually come away feeling even more confident because that's what this is about.

It's about confidence developing, I think.

One thought on that is like, it's possible to stay purely internal to math.

And that's the internal logic of proofs.

It's also possible to stay internal to general claims.

And do philosophy.

Is it possible to stay purely within the realm of empiricism?

In the trivial case, yes.

If you're just aggregating data sets and never interpreting them, yes.

But empirical in application must either connect to a general question.

Like, you know, again, we observed the ants and now we're going to talk about evolution.

So that's still qualitative, but it's an edge.

And then it rests on math when it's more statistical and on generalities when it's more conceptual.

But it's just sort of interesting that the internal logic of empiricism is basically just data integrity.

But it's like pre-interpretation, which is perfect for it being X.

Whereas there's a whole guild structure and discipline around internal mathematics and internal generalizations and philosophy.

And so it's like, and I also really agree with that, having them, I thought you were going to say, just get them in the same room and then just talk about food or music or something like that.

No, but I know what you mean.

Have them and just like, we're all learning by doing in this space.

And to say that this is a lifelong journey would just be a total understatement.

It's broader and deeper than that.

So how do we recognize the complexity of this arrangement as well as the adjacent possibilities, other ways that it could have been made, other end states that we could have reached?

Bringing in any number of fourth points to make a tetrahedra?

Any other number of orders of paths that we could have done to reach this point?

Like if we had started with these little dots and then later added the arrows and how did we start this whole thing?

With examples of each edge.

So getting to the super like fractal multi-scale level where, and we didn't even add any formalisms or mathematics here.

But we could have done some pseudo code or some pseudo equations to really start to link this and to lean more on our mathematical leg.

And that's the whole thing that you just said about confidence, Dean, which is like, this is the space that we're going to be bouncing around.

And you can ignore statistics and you can just have an emotional response to p-values and stuff like that.

But you will need that as part of the journey.

We can't just be like the post office vehicles that don't make left turns.

We have to make all of our tools available at the team level.

And then trust that with like distributed cognition and with guides and wayfinders and peer facilitation that like we can make it work despite how vast this area is.

So yes, Stephen, as we go for our final commentary.

Yeah, I think you made a good point there.

With that green arrow, just one last thing with the green arrow you put there, could that be more implementable?

Like it becomes more implementable as you go that direction away from the math with a mixture of general and empirical?

Just curious.

So the arrows, yeah, the arrows, I mean, again, they're open to interpretation and other ways it could have been done.

I was thinking what links up math and empirical?

They're both more numerical.

That's the bottom edge.

Okay, then what makes math and general similar?

They're less example-like.

And then what is the similarity between empirical and general?

So what is not just pointing, what's the arrow that doesn't point to general?

What's the arrow that points between general and empirical?

And what distinguishes mathematics from general and empirical?

It might be its formalism, but general is it.

Well, I'm thinking implementation, or maybe this does get into teleology.

Like it shows you more, like the empirical is what's actually happened.

So like I always, this is an example, is the accountant said to us once, and it always stuck in my mind is, you know, I've said this before.

Or what is an organization doing?

What's it about?

And he said, don't tell me your mission.

Don't tell me your vision.

Show me your budget.

Show me what you're doing.

Or show me where you spend your time.

So in a way it's like that.

What's the empirical as well as the, that goes to the general to show me, well, that's, that gives an idea then of something that can be done as opposed to words or even a mathematical,

well, words can be like a mathematical formula in many ways to help tell people what you want them to think, but maybe not what is going on.

So anyway, I like those three arrows.

I think it's pretty useful.

Any final comments in the last one minute?

Dean, and then Lou.

So real quickly, I think that a lot of times conferences people go and they want to talk about what they know.

But I think that this example today was an example of where you come in, not necessarily knowing, but discovering.

I think it's another fine example of what we're talking about when we're talking about active inference.

If I, if I caught Stephen Mann and Connor in the same room, the place that I would want to begin is what, what don't you know?

How would you guys attack this from not your strength preferences, probabilities and fit, but from what you, what you don't know, but is still in the room.

And I think that's very complimentary to the, to the more traditional conference way of disseminating information.

There's my word for today.

And then even socially at the conference, people connect with the people who they do know.

And that's the little click network instead of connecting with people who we don't know.

And so it also ties to hopefully some of our values too.

So blue, any last thoughts?

Just, I'm just thought of complexity science and the interdisciplinarity and the real value that goes into co-learning or working outside of your comfort zone.

Because really like that's where, where the growth happens.

So I like those ideas a lot.

Yep.

Working with people, you know, can have amazing pragmatic utility and it must be balanced with the epistemic and with expanding our horizons and perspectives as well.

So yeah, totally agreed.

Well, another fun one.

Can't say we prepared too much, but I think in the end it went just fine.

So thank you all for joining and for participating and we'll...